Sivesh PB

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Education

CHRIST (Deemed to be University) — Bangalore, Karnataka
M.Sc. in Artificial Intelligence and Machine Learning — *July 2024 – Present*

Yeldo Mar Baselios College (Affiliated to Mahatma Gandhi University) — Ernakulam, Kerala
Bachelor of Computer Applications — *August 2021 – May 2024*

Experience

Ideaelan — Remote
Junior AI Engineer — *January 2025 – Present*

- Built a **Monday.com-style project and ticket management platform** in-house for **Infinity X**, replicating core workflow functionality and extending it with **Agentic AI automation**.
- Built **Flask-based LLM microservices** integrated with **LangChain** agentic pipelines and **RAG** retrieval (ChromaDB), enabling real-time NLP inference for production support workflows.
- Implemented **embedding-based related-ticket detection** using vector similarity search, surfacing duplicate/linked tickets and reducing redundant agent effort at scale.
- Deployed and maintained AI services with **Docker containerisation** and **Git-based CI/CD**, covering model iteration through to production release.
- Built **AI-powered reply suggestion** using LLMs to generate context-aware draft responses, improving resolution consistency across the support team.
- Owned the **end-to-end system architecture** with **Azure Cosmos DB**, including API design and backend services.

UIAI Technologies — Chennai, Tamil Nadu
GenAI Intern — *March 2025 – July 2025*

- Engineered autonomous **multi-agent AI workflows** using **AGNO and LangChain**, and built **RAG pipelines** with vector databases, improving knowledge retrieval accuracy by **~35%**.
- Developed and deployed **Flask-based LLM services** integrated with **Ollama** and vector similarity search, enabling real-time NLP for internal tools, following **Git-based CI/CD workflows**.
- Optimised and fine-tuned LLM models for hiring automation, increasing candidate screening precision by **25%** and reducing manual review time by **40%** — a production-grade AI solution delivering measurable business impact.

Projects

Multi-Agent AI Attrition Analysis System | *LangChain, Python, Streamlit, Docker, RAG*

- Engineered a modular multi-agent architecture using **LangChain** with dedicated agents for data processing, statistical analysis, and ML-based attrition prediction, achieving **65% faster** HR insight generation.
- Implemented a **RAG pipeline** with ChromaDB for context-aware querying over employee records, enabling natural language HR analytics without manual report generation.
- Containerised the full system with **Docker** and deployed an interactive **Streamlit** dashboard, making complex workforce analytics accessible to non-technical HR stakeholders.

Career Placement Assistant AI Agent | *Python, Streamlit, RAG, Groq API, ChromaDB*

- Built an interactive web-based AI assistant to provide real-time analytics on **3,000+** university placement records, covering salary trends, company-wise hiring patterns, and branch-level insights.
- Implemented a **Retrieval-Augmented Generation (RAG)** backend with **ChromaDB** as the vector store, achieving **92% retrieval accuracy** for context-aware, data-grounded responses.
- Designed a responsive **Streamlit** UI capable of handling complex multi-part queries on placement statistics, significantly reducing the time counsellors spend on manual data lookup.

Mental Health Risk Detection & Response Agent | *RoBERTa, Mistral, Streamlit, Ollama*

- Fine-tuned a **RoBERTa** transformer model on Reddit mental health corpora, achieving **88% accuracy** on multi-class sentiment and risk-level classification.

- Integrated **Mistral via Ollama** for local, privacy-preserving generation of context-aware empathetic responses, ensuring no sensitive user data leaves the device.
- Deployed an end-to-end **Streamlit** interface combining real-time risk classification with LLM-driven support messaging, suitable for early mental health intervention use cases.

Automated HR Management System | *Python, Flask, PostgreSQL, React, Docker*

- Designed and built a production-grade **full-stack HR web application** with a secure **Flask** REST API backend and a **React** front-end, supporting end-to-end employee lifecycle management.
- Architected a **PostgreSQL** database schema managing records for **200+ employees**, optimised with indexing and normalisation for performance and scalability.
- Containerised the application using **Docker**, ensuring consistent deployment and seamless integration between the client, server, and database layers.

Achievements

Conference Paper (Presenter) – *Leveraging GenAI Foundation Models for Plant Disease Detection: The AgriCLIP-CNN Architecture*, International Conference on Artificial Intelligence (ICAI).

Journal Submission (Q1) – *Advances in Deep Learning for Plant Phenotyping through Multimodal Image Fusion: A Critical Review for Precision Agriculture*, submitted to a Q1-indexed journal (under review).

Winner – **Best Innovation Award**, SRM University Hackathon for an AI-powered sustainability solution.

Technical Skills

Languages: Python, SQL, Java, C, JavaScript

Generative AI & Agentic AI: LangChain, LangGraph, LangSmith, LlamaIndex, RAG, AI Agents, Multi-Agent Systems, CrewAI, AutoGen, Fine-tuning, LoRA/QLoRA, RLHF, Prompt & Context Engineering, SLMs, OpenAI API, Anthropic API, Groq

Vector Databases & Embeddings: Chroma, Weaviate, Qdrant, Embeddings, Semantic Similarity Search

AI Engineering & MLOps: Model Deployment, Model Monitoring & Evaluation, Docker, Git, CI/CD, MLflow, Weights & Biases, Model Versioning, Data Pipelines

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Sentence-BERT, OpenCV, LSTM, CNN, RNN, ARIMA, Prophet, GAN, GNN, Autoencoders, XGBoost

Data & Engineering: Pandas, NumPy, Polars, ETL/ELT, Data Lakes, Data Warehouses, REST API, MySQL, MongoDB, Firebase, Cosmos DB

Visualization & Web: Matplotlib, Seaborn, Plotly, Power BI, Tableau, Streamlit, Gradio, Flask, Django

Cloud: AWS, GCP

Concepts: Supervised/Unsupervised Learning, Reinforcement Learning, Transfer Learning, Few-Shot Learning, SHAP, LIME, BLEU/ROUGE, ROC-AUC, Agent Design, Microservices

Tools: Git, VSCode, Jupyter, Google Colab, Excel